

ARbnb: An AR Indoor Navigation and Spatial Annotation System

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ABSTRACT

ARbnb is an augmented reality (AR) navigation and spatial annotation system designed to help Airbnb guests orient themselves within unfamiliar rental properties. Using Unity 6 and AR Foundation 6.4, the system overlays real-time directional waypoints, host-authored annotations anchored to physical objects, and a progressive discovery map directly onto the guest's camera view. Hosts pre-author location-specific notes — such as appliance instructions or house rules, which persisted across sessions using Google Cloud Anchors. The system targets all the ARCore and ARKit compatible devices which doesn't require dedicated depth sensors and enhances experiences on LiDAR-equipped hardware. Our application aims to reduce guest to host support friction and improve spatial familiarity in short-term rental environments through contextual, spatially aware AR guidance.

1. INTRODUCTION AND MOTIVATION

Guests arriving at an unfamiliar Airbnb rental face a recurring orientation problem: locating appliances, understanding house rules, and navigating an unknown layout - all without immediate host assistance. Current solutions rely on printed manuals, PDF guides, or host messaging, none of which are spatially contextual or interactive. Augmented reality (AR) presents a natural solution: overlaying digital information directly onto the physical space allows guests to receive guidance exactly where and when it is relevant.

ARbnb addresses this by anchoring host-authored notes and navigation cues to the real world, creating a persistent, personalized spatial memory of the home that activates on guest arrival and clears automatically on checkout, preserving guest privacy.

2. RELATED TECHNOLOGIES AND PRIOR WORK

Indoor AR navigation has been explored in museum and hospital contexts. This typically relies on marker-based tracking or pre-mapped environments. Recent advances in markerless SLAM - through ARKit and ARCore - have enabled device-agnostic spatial understanding without dedicated infrastructure. Google Cloud Anchors extends this to cross-device, persistent experiences that survive across sessions and users.

Prior work on AR annotation systems such as Studierstube demonstrated the value of spatially anchored information layers, though none target short-term residential rental contexts. ARbnb combines persistent cloud anchors with a host-facing authoring workflow, distinguishing it from existing navigation and annotation systems.

3. PROPOSED SYSTEM

The system comprises two components. The **guest mobile application**, built in Unity 6 with AR Foundation 6.4, provides: (i) floor-plane detection and spatial mapping on first walkthrough; (ii) floating AR waypoint arrows routing guests to requested destinations; (iii) proximity-triggered annotation panels anchored to objects such as appliances or entry points; and (iv) a screen-space discovery HUD revealing rooms as guests explore.

The **host web dashboard**, built in React and Three.js, allows hosts to place annotation pins on an uploaded floor plan and

author text notes, stored in a Node.js/PostgreSQL backend and synchronized to guest devices via Google Cloud Anchors. The system targets all ARCore-supported Android devices (2019+) and ARKit-compatible iOS devices (iPhone 12+), with LiDAR used as an enhancement where available.

4. IMPLEMENTATION PLAN

Foundation. Unity 6.4 project setup, AR Foundation configuration for Android and iOS, floor plane detection, and a floating AR welcome label rendered in world space.

Anchor System. Local anchor persistence via ARAnchorManager, followed by cloud sync using ARCore Extensions and the Google Cloud Anchors API.

Annotation Pipeline. REST API connecting Unity to a Node.js backend; world-space annotation UI panels with proximity-based visibility culling (max 3 visible simultaneously).

Host Dashboard and Wayfinding. React/Three.js dashboard for floor plan upload and pin authoring; AR waypoint arrow system for room-to-room navigation derived from the floor plan graph.

Discovery Map and Polish. Screen-space discovery HUD, offline caching of all annotation data, UI refinement, and end-to-end testing on Samsung Galaxy S25 Ultra (Android 15, ARCore).

5. TEAM RESPONSIBILITIES

Team Name: Musashi

Diksha Chottani - Unity AR lead: AR Foundation setup, spatial anchor system, wayfinding arrow navigation and Google Cloud Anchors integration

Jayananden Meenakchisundaram - Backend and infrastructure lead: Node.js REST API, PostgreSQL/PostGIS spatial schema, Supabase authentication and WebSocket synchronisation

Kartik Singhal - Frontend and UX lead: annotation UI panels, discovery HUD, React/Three.js host dashboard and floor plan pin authoring workflow